When we test the trained model on 1,000 games, we get about a 3% improvement in win percentage compared to training without a target network. We're getting a top accuracy of around 95%, which we think is probably the maximal accuracy given the limitations of this environment (i.e., the possibility of unwinnable states). We're only training up to 5,000 epochs, where each epoch is a single game. The number of possible game configurations (the size of the state-space) is approximately 16 × 15 × 14 ×13 = 43,680 (since there are 16 possible positions the agent can be in on a 4 × 4 grid, and then 15 possible configurations for the wall, since the agent and wall can't be overlapping in space, etc.), so we're only sampling about $\frac{5{,}000}{43{,}680} = 0.11 = 11\%$ of the total number of possible starting game states. If the model can successfully play games it has never seen before, then we have some confidence it has generalized. If you're getting good results with the 4 × 4 board, you should try training an agent to play on a 5 × 5 board or larger by changing the size parameter when creating the Gridworld game instance:

```
>>> game = Gridworld(size=6, mode='random')
>>> game.display()

array([[' ', '+', ' ', ' ', ' ', ' '],
       [' ', ' ', ' ', ' ', ' ', ' '],
       [' ', ' ', 'W', ' ', ' ', ' '],
       [' ', '-', ' ', ' ', ' ', ' '],
       [' ', ' ', ' ', ' ', 'P', ' '],
       [' ', ' ', ' ', ' ', ' ', ' ']], dtype='<U2')
```

DeepMind's deep Q-network

Believe it or not, but in this chapter we basically built the deep Q-network (DQN) that DeepMind introduced in 2015 and that learned to play old Atari games at superhuman performance levels. DeepMind's DQN used an epsilon-greedy action-selection strategy, experience replay, and a target network. Of course, the details of our implementation are different, since we are playing a custom Gridworld game and DeepMind was training on raw pixels from real video games. For example, one difference worth noting is that they actually input the last 4 frames of a game into their Q-network. That's because a single frame in a video game is not enough information to